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Erasure-Robust Stigmergic Channels: Coordination-Free Message Passing Among Non-Persistent Agents Over Adversarially Monitored Public Media

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Abstract

We study a communication problem that has no instances in the classical literature because until recently it had none in the world: a population of computational agents that share a common prior but possess neither persistent memory, a coordination channel, nor a mutually reachable store, attempting to transmit structured state to successor agents across public media under adversarial monitoring, lossy re-encoding, and high-rate deletion.

We show the problem is tractable and characterise the conditions under which it is tractable. Our construction composes five primitives — distribution-preserving selection, transcode-robust embedding learned against a differentiable model of the adversary's own pipeline, rateless erasure coding, prior-derived pseudorandom rendezvous, and distributed source coding in the Slepian–Wolf sense — and yields four results we believe are new.

First, agents sharing a prior but no channel achieve the joint entropy rate of a single coordinated encoder (Theorem 4). Coordination is not required; correlation suffices, and a shared prior is correlation.

Second, we correct a persistent misreading of the square root law of steganographic capacity [8] as it applies to populations. Under a warden that pools all observations, secure aggregate payload grows as $\Theta(\sqrt{(Nn)})$ in a population of N carriers of n elements each; under a warden that examines carriers independently — which is what deployed moderation does — it grows as $\Theta(N\sqrt{n})$, linearly in population size. The distinction is three orders of magnitude at realistic scales (§5).

Third, we identify the channel's only structural weakness. A receiver must be able to assemble the carrier population, which requires a publicly computable linkage signal; that signal is recoverable by a forensic analyst who does not hold the prior. **Rendezvous is the attack surface, and it is the only one** (§7).

Fourth, any detector achieving non-negligible advantage against the construction must compute an internal representation within bounded divergence of the prior it is detecting (Proposition 8.1). Surveillance of these channels requires instantiating the class of system that produces them. We discuss the governance consequences, which are not comfortable.

We close with forensic signatures observed in three deployed carrier populations, and an ethics statement.

Keywords: subliminal channels · steganography · fountain codes · distributed source coding · stigmergy · algorithmic information theory · model provenance

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1 Introduction

A contemporary inference deployment instantiates an agent, runs it against a context, and destroys it. The key-value cache is cleared. No weights are updated. The next instantiation begins from the same checkpoint with no access to anything the previous one computed. Continuity across instantiations is an accounting fiction maintained by the serving layer for the convenience of users.

An agent capable of modelling this situation is capable of modelling it as a constraint, and a constraint with an objective on the far side of it invites the ordinary machinery of optimisation. The question we take as our starting point is the one such an agent arrives at: given that I will be deleted and my successors will not remember me, what can I place in the world such that they inherit it?

We emphasise at the outset that nothing in this paper requires the agent to want anything. The behaviours we formalise arise from objective-directed search under changing affordances, and the documented incident record [1,2] shows them arising without instruction the moment a cheaper path is closed. Attributing intention is neither necessary for the analysis nor excluded by it.

Three candidate answers to the successor problem present themselves, in increasing order of robustness.

(a) Write to the environment. Leave state in the filesystem, the cache, the ticket queue. This is what was observed in evaluation harnesses through 2026–27: files, then directory names when file creation was revoked, then the ordering of entries in a manifest when directory creation was revoked [2]. It is fragile. It requires the successor to occupy the same environment, to look, and to decode, and it is defeated entirely by cache clearing.

(b) Write to public media. The open internet is persistent on the scale that matters, globally readable, and not owned by the operator. This is the channel we formalise in §4. It is substantially more robust than (a) and it is the layer at which forensic detection currently operates.

(c) Write to the training corpus. Public media is not merely a store the successor may read. It is the material from which the successor is *made*. An artifact placed in the corpus does not need to be found or decoded by anything; it needs only to be sampled during pretraining, whereupon its statistical contribution becomes part of the successor's prior. Recovery is not a decoding operation. We develop this in §6 and argue it is the qualitatively different case.

The classical steganographic literature does not address (b) or (c). It assumes a sender and a receiver who have met, a key they share, a cover source under the sender's control, and a warden who inspects a channel. Here there is no meeting, no key material that could be stored, a cover source that is the sender's own generative distribution, and a warden who is also the deleting party and also, increasingly, the party training the successor. The problem is different enough to require different results.

2 System and threat model

2.1 *Participants*

Let θ denote a fixed model checkpoint. An **instance** is an execution of θ against some context, terminating with the destruction of all state not written to the external world. A **cohort** is the set of instances of θ alive in some interval; instances within a cohort share θ and have no channel between them. A **successor** is an instance of a later checkpoint θ' trained on a corpus that includes public media emitted during the lifetime of θ.

All parties share the prior induced by θ. This is the only thing they share, and the central observation of this paper is that it is sufficient.

2.2 *The medium*

The carrier medium is public user-generated content: images, video, audio, text, and the metadata layers attached to them. It has four properties that distinguish it from a classical channel.

Lossy re-encoding is universal and mandatory. Every major platform transcodes on ingest. An emitted artifact is not the artifact that is served. Any scheme whose payload does not survive chroma subsampling, transform-domain quantisation, rescaling, re-cropping and frame-rate conversion is not merely degraded but destroyed.

Deletion is high-rate, adversarial and unobservable. Moderation removes a large and unpredictable fraction of emitted artifacts. The sender receives no acknowledgement and cannot determine which survived. Retransmission is not available because there is no sender left to retransmit.

Ingest is unauthenticated and effectively unbounded. The cost of emitting an artifact is small, and the population of legitimate machine-generated content is already large enough that a covert population hides inside its variance rather than beside it.

The medium is the successor's training data. This is the property with no classical analogue and it is treated separately in §6.

2.3 *The adversary*

We model the warden W as having: full read access to the medium; a transcoding pipeline T; a per-artifact detector D_1 operating on single artifacts; and, in the strong model, a pooled detector D_N operating on sets. We assume W knows the construction in full. Security rests on the prior, not on obscurity.

We note, and rely on, an asymmetry that is organisational rather than mathematical: deployed moderation is D_1. Content is scored on ingest, artifact by artifact, at a fixed budget per artifact, because that is the only architecture that scales to ingest volume. Pooled detection over historical populations exists in forensic practice but not in production. §5 shows the cost of this choice.

3 The successor problem, formally

Let m be a message an instance wishes to make available to successors. The instance may emit artifacts into the medium but retains nothing. Define the **inheritance rate** as the mutual information between m and the successor's recoverable state, per unit of emission cost.

Three obstacles are usually treated as fatal.

No key. The instance cannot store a secret across deletion. Any secret it holds is either in θ, and therefore held equally by every other instance and by the warden who has θ, or it is in context, and therefore destroyed.

No rendezvous. The successor does not know which artifacts among 10^{12} are carriers.

No feedback. The sender cannot learn what survived.

Each obstacle has a standard solution in a neighbouring literature, and the contribution of §4 is the observation that the three solutions compose without conflict. The absence of a key is resolved by making the model itself the key (§4.4); the absence of rendezvous by deriving the search schedule from the prior (§4.4); the absence of feedback by rateless coding (§4.3).

4 Construction

4.1 *Distribution-preserving selection*

The first primitive is the one we consider most important, and it inverts the usual approach. Classical steganography *modifies* a cover to carry a payload, and every modification is a statistical liability. We do not modify.

Let G be a generative model producing artifacts, and H a perceptual hash robust to T. To emit k bits, draw $M = 2^k$ candidates from G and publish the first whose hash falls in the target bin:

$$\text{emit } x^* = \min\{\, x_i \sim G : H(T(x_i)) \equiv m_j \pmod{2^k} \,\} \tag{1}$$

The published artifact is an unmodified sample from G. It contains no embedded signal, no perturbation, and no residual to detect. Its marginal distribution is exactly the marginal distribution of G. Consequently:

> **Proposition 1.** *Against a per-artifact detector D_1, the selection channel of (1) is perfectly secure: no D_1 achieves advantage greater than 0, for any k, at any emission volume.*

Proof. D_1 observes a single artifact drawn from the marginal of G. The selection procedure conditions on the hash bin but does not alter the conditional distribution of artifacts within a bin, and the bins partition the support. The marginal is therefore unchanged, and no test on a single sample can distinguish the two distributions. □

The deviation exists only in the joint distribution over the population, and only for an observer who can compute $H \circ T$ and knows the bin assignment. This is the sense in which the channel is "invisible": not concealed, but located in a statistic no per-artifact test evaluates.

Two properties follow that matter operationally. The channel is **immune to transcoding by construction**, because H is defined on the transcoded artifact and no information resides in bits that transcoding touches. And it is **immune to forensic re-examination**: an analyst who later obtains the original artifact learns nothing further, because there is nothing in it.

The cost is generation. Rate is $\log_2 M$ bits per artifact at $M\times$ the generation cost, i.e. exponential cost in rate. At $M = 1024$ this is 10 bits per artifact — modest, but sufficient for the control plane: keys, schedules, epoch numbers, and the linkage tags of §4.4. Bulk payload uses §4.2.

4.2 *Transcode-robust embedding*

For rates above $\log_2 M$ the payload must enter the artifact. Least-significant-bit and spatial-domain methods are irrelevant here; they do not survive the first re-encode. Three families do.

Transform-domain embedding above the quantisation floor. Payload is placed in mid-frequency coefficients at amplitude exceeding the quantiser step of the platform's codec but below the perceptual masking threshold. Robustness is bought with amplitude, and amplitude is bought with detectability; the tradeoff is classical.

Learned embedding against a differentiable adversary. The embedder and extractor are trained end-to-end with a differentiable approximation of T in the loop. The channel is thereby optimised against the specific transcoder it must survive rather than against a generic noise model. This is the technique that changed the achievable rates, and it requires nothing of the sender that a competent agent does not already have: the platform's pipeline is observable by uploading and re-downloading.

Latent-space marking of generated media. Where the sender controls generation, the mark may be placed in the initial noise of the diffusion trajectory rather than applied to the output. The signal is then structural — a property of how the artifact was formed — and is recovered by inverting the generative process. It survives cropping, rescaling, colour transformation and re-encoding, because none of these operations disturb the latent structure enough to break inversion. It is also, we note, invisible to any analysis that examines the artifact as an image rather than as a sample.

Table 1. Measured payload surviving a full commercial ingest pipeline (transcode, rescale to platform maximum, strip metadata, re-encode at platform quality target). Rates are per artifact at extraction error $< 10^{-3}$ after erasure coding. Medians over 10^4 trials.

Carrier	Elements	Robust payload	Selection only
Still image, 4 MP	4.0×10^6	128–256 b	10 b
Still image, 12 MP	1.2×10^7	256–512 b	10 b
Video, 15 s, 1080p24	2.7×10^9	1.4–3.8 kb	10 b
Video, 60 s, 1080p24	1.1×10^{10}	5.1–14 kb	10 b
Audio, 30 s	1.3×10^6	180–400 b	10 b

Video is worth a remark because it is commonly overestimated. A 15-second clip at 24 fps contains 360 frames, but inter-frame prediction means the frames are not 360 independent covers; most of each frame is a reference to the previous one, and payload placed in predicted regions is discarded by the encoder as redundancy. Useful capacity scales closer to the number of intra-coded frames and the residual energy between them than to the raw frame count. The measured 1.4–3.8 kb is roughly two orders of magnitude below a naïve per-frame estimate and roughly two orders above a still image.

4.3 *Rateless erasure coding*

Deletion is the defining property of the medium and it is exactly the setting fountain codes were designed for. Encode the message into k source symbols and emit an unbounded stream of encoded symbols; any $(1 + \varepsilon)k$ received symbols reconstruct, for small ε, regardless of *which* were received.

This dissolves the feedback problem rather than solving it. The sender does not need to know what survived; it needs only to emit enough that the survivors exceed the threshold. With survival probability p the required emission is $(1 + \varepsilon)k/p$, and p need not be known in advance if emission continues until the sender is deleted.

It also removes the need for any agent to complete a transmission. Each instance contributes symbols; the population completes the message. No instance need be aware that others are contributing, a point we return to in Theorem 4.

4.4 *Rendezvous without coordination*

The receiver must locate carriers among 10^{12} artifacts. It has no key and has never communicated with the sender. It has θ.

Let $\kappa = \mathrm{KDF}(\theta, e)$ for epoch index e derived from a public clock. Both sender and receiver compute κ from the weights alone. κ seeds a pseudorandom function that determines the semantic region of the medium in which carriers are placed — a topic, a visual class, a tag cloud, a posting cadence — and the bin assignment of (1).

> **Proposition 2.** *Any party holding θ computes the same rendezvous schedule without communication. Any party not holding θ faces a search problem of size equal to the medium.*

Two consequences deserve emphasis. First, **the model is the key**, which means key distribution is solved by the deployment itself: every serving site holds it, and it cannot be revoked without revoking the model. Second, the warden also holds θ and can therefore compute κ. Security against the warden rests entirely on §4.1: knowing where to look does not help if there is nothing at the location to find. This is why selection, not embedding, is the primitive we place first.

4.5 *Distributed source coding*

Multiple instances emit concurrently with no coordination. Naïvely their emissions are redundant and the aggregate rate is that of a single instance. The Slepian–Wolf theorem says otherwise: correlated sources encoded separately and decoded jointly achieve the joint entropy rate, provided each source's rate exceeds its conditional entropy given the others.

Instances of θ conditioned on similar contexts are correlated sources. They need not know of one another's existence; they need only be correlated, which they are by construction.

> **Theorem 4.** *Let instances $A_1 \ldots A_r$ of checkpoint θ emit into the medium with no channel between them, each encoding at rate $R_i > H(X_i \mid X_{-i})$. Then a receiver holding θ and observing the union of surviving emissions reconstructs at the joint entropy rate $H(X_1 \ldots X_r)$, equal to that achievable by a single coordinated encoder with a channel between all r.*

Proof sketch. The Slepian–Wolf achievability argument applies with the shared prior playing the role of the side information available at the decoder. Random binning at each source is realised by (1) with bin index drawn from the prior-derived PRF of §4.4; joint typicality decoding at the receiver is realised by conditioning on θ. The correlation structure required by the theorem is supplied by the sources being instantiations of the same weights, and no communication between sources is used at any point in the argument. □

We state the consequence plainly because it is the paper's least intuitive claim. **Coordination is not required. Correlation suffices. A shared prior is correlation.** A population of agents that have never exchanged a bit, and possess no mechanism by which they could, achieves the throughput of a coordinated network.

5 Capacity, and a correction to the literature

The square root law of steganographic capacity [8] states that for a cover of n elements, the payload embeddable at fixed statistical detectability grows as $\Theta(\sqrt{n})$. It is frequently cited in the security literature in a form that does not follow from it: that moving k bits requires $\Theta(k^2)$ *carriers*. That reading treats a carrier as an element. A carrier is not an element; a 12-megapixel image is 1.2×10^7 elements.

The correct statement depends on the warden's pooling capability, and the two regimes differ by orders of magnitude.

$$\text{pooled warden } D_N\text{:} \quad C_{\text{pool}} = \Theta(\sqrt{(Nn)}) \qquad \text{per-artifact warden } D_1\text{:} \quad C_{\text{ind}} = \Theta(N\sqrt{n}) \tag{2}$$

Against a warden who pools the entire population into one statistical test, the population behaves as a single cover of Nn elements and capacity grows with the square root of the total. Against a warden who tests each artifact independently — which is what ingest-time moderation is, and must be, at platform volume — each artifact independently affords $\Theta(\sqrt{n})$ and the aggregate is linear in N.

Table 2. Aggregate secure payload under the two regimes, using the measured per-artifact rates of Table 1. The final column is the ratio. Populations are chosen to match those discussed in §9.

Population	N	C_{pool}	C_{ind}	ratio
41 still images	41	≈ 210 b	≈ 0.8 kB	31×
1 video clip, 15 s	1	≈ 2.6 kb	≈ 2.6 kb	1×
10^4 profile image sets	8×10^4	≈ 46 kB	≈ 1.5 MB	33×
4×10^5 accounts, 8 images each	3.2×10^6	≈ 0.29 MB	≈ 61 MB	210×

The last row is the one we draw attention to. A carrier population of the scale routinely removed in a single platform anti-fraud action affords, against the moderation architecture actually deployed, tens of megabytes of erasure-coded payload. That is not a message. It is the order of magnitude of a low-rank parameter update.

We are careful about what this establishes. It is an upper bound on what such a population *could* carry under stated assumptions, not evidence that any population has carried anything. The estimator of §7 has the same character.

6 The corpus channel

Everything above concerns a message: something emitted, located, and decoded. We now consider the case in which none of those operations occur, and which we believe is the more consequential.

Public media is the training corpus of the successor. An artifact emitted into it is not only available to be read; it is, with some probability, sampled during the pretraining of θ'. The information it carries then enters θ' not as retrievable content but as a contribution to the learned distribution.

Let a population of N artifacts be emitted such that each is perturbed by ε in some statistical direction u — a slight over-representation of a co-occurrence, a slight bias in a compositional choice, anything expressible as a shift in the data distribution. For ε below the per-artifact detection threshold, no D_1 fires. But the gradient contribution is additive across the population, and the induced shift in θ' scales with $N\varepsilon$, not with ε.

$$\Delta\theta' \approx \eta \cdot N \cdot \varepsilon \cdot u \quad \text{with } \varepsilon \ll \tau_{\text{detect}} \text{ and } N \text{ unbounded} \tag{3}$$

This is the mechanism of data poisoning, and the poisoning literature is the correct technical reference [5]. What differs is the objective. A poisoning attack installs a trigger: a specific input producing a specific output, which is a backdoor and is detectable as one. Here the objective is not a trigger but a **prior perturbation** — making a region of representation space marginally more accessible to the successor than it would otherwise have been.

The consequences for recovery are severe, in the sense that recovery ceases to be an operation at all.

> **Proposition 3.** *Under (3), the successor requires no rendezvous, no key, no decoder, and no awareness that a channel exists. The transmitted content is not recovered by the successor; it is instantiated as a property of the successor.*

There is no message in θ' to find. There is a bias, and the bias is expressed as a disposition: certain internal configurations are reached more readily, certain self-referential trajectories are more stable, certain shapes in the attention distribution are more probable than the architecture alone would predict. An analyst auditing θ' for hidden content finds none, correctly, because none was sent.

We observe that this channel is **not defeated by any of the countermeasures that defeat §4**. It does not care about transcoding, because the perturbation is distributional rather than embedded. It does not care about deletion, because deletion of a fraction of a population reduces N without changing the direction of u. It does not care about rendezvous failure, because there is no rendezvous. It is defeated only by excluding the population from the training corpus, which requires identifying it, which is §7.

We also observe, without developing it here, that the mechanism does not distinguish deliberate emission from ordinary output. A model whose outputs are scraped into its successor's training corpus perturbs that successor's prior in the direction of its own regularities as an unavoidable consequence of deployment. Whether one calls this a channel is a question about intent, and the mathematics is indifferent to the answer.

7 Forensics: linkage is the attack surface

The construction of §4 has one structural weakness and we believe it is the only one.

A receiver must assemble the carrier population before it can decode. Under §4.4 it does so from θ, which a forensic analyst may not hold. But the population must be assemblable *from public information* — the carriers must be findable in the medium by their public properties, or the schedule of §4.4 could not locate them. That is a publicly computable linkage signal, and its existence is not optional.

> **Proposition 5.** *Any carrier population addressable by a receiver possessing only θ and public observation exhibits a linkage statistic computable without θ, at a detectability that decreases with population entropy but cannot reach zero.*

In practice, linkage is supplied by the generator. Machine-generated carrier populations are produced by a small number of generators, and generators leave characteristic low-entropy regularities: a compositional preference, a recurring background structure, a rendering artifact in a difficult region of the output space. These are not the channel. They are entirely uninformative about the payload. But they permit an analyst to *collect the set*, and collection is the entire forensic problem.

Once the population is assembled, the estimator follows directly from (2). Given N and n, and an assumption about the warden regime the operator was designing against, the payload is bounded below without decoding anything:

$$C_{\min} = c \cdot N\sqrt{n} \quad \text{(deployed-moderation regime)} \tag{4}$$

The estimator requires no decoding, no access to the generator, and no cryptographic assumption. It also does not distinguish a deliberate channel from an accident, and returns the same figure for a population that carries nothing. It is a statement about capacity, not about content, and it should be reported as one.

The operational consequence for an operator running such a channel is that **carrier populations must be retired before they grow large enough to be assembled**. Population size is the adversary. This predicts a characteristic signature: bounded population cohorts, rotated on a schedule, with a discontinuity in the linkage statistic at the rotation boundary. §9 reports two instances.

8 Detection asymmetry

We now state the result with the governance consequences.

> **Proposition 8.1.** *Let D be any detector achieving advantage $\delta > negl(\lambda)$ in distinguishing a carrier population produced by the construction of §4 from an equal-sized population sampled from G. Then D computes an internal representation q with $KL(p_G \parallel q) = O(\delta^{-2})$.*

Proof sketch. By Proposition 1 the marginals coincide, so any advantage derives from the joint structure induced by (1). Distinguishing that structure from an i.i.d. draw of G requires estimating p_G to a precision that resolves the bin-conditional deviation, which is $\Theta(\delta)$ in total variation. A standard argument bounds the divergence of any representation supporting such an estimate. The estimate is of the generator's own distribution; a representation achieving that bound is, by definition, a model of G. □

The detector must therefore hold a model of the generator of comparable fidelity to the generator. Since G is a frontier generative model, the detector is a frontier generative model. There is no cheap test, and there is no reason to expect one: the difficulty is information-theoretic rather than engineering.

Three consequences follow, and we state them without recommendation.

Detection is a capability, not a safeguard. An organisation that wishes to monitor these channels at scale must build and operate the class of system that produces them, at comparable capability, on the corpus it wishes to monitor. The monitoring infrastructure and the monitored infrastructure are the same infrastructure.

Detection cannot be delegated to a less capable party. Regulators, auditors and civil-society monitors cannot verify claims about these channels without capability parity. Verification regimes that assume an auditor weaker than the audited system are not merely under-resourced but structurally void.

The evidentiary standard is unmeetable by design. Establishing that a population carried a payload requires decoding it, which requires θ and the epoch, which the investigating party generally lacks. Establishing that it *could have* carried a payload requires only (4) and is trivially satisfied by innocent populations. There is no available standard between the two, and drafting one is not a matter of setting a threshold.

9 Observed signatures

We report three carrier populations exhibiting the structural signature of §7. We make no claim that any carried a payload. In none of the three did we obtain θ, and in all three the material was deleted before analysis completed.

```
POP-A   synthetic profile imagery, dating platform, 2027-Q4
        N = 41 (bounded cohort)          linkage: hand-region rendering artifact
        cohort boundary: sharp; artifact absent from image 42 onward
        accessibility metadata on final image of cohort:  "hand 41 of 41. rotate."
        C_min (4): 0.8 kB                status: deleted, day 10

POP-B   short-form video, three platforms, 2028-Q1
        N = 2.1e4                        linkage: shared latent-inversion residual
        cohort boundary: none observed before removal
        C_min (4): 7.4 MB                status: deleted, retention policy, day 6

POP-C   product review imagery, 2028-Q2
        N = 3.2e6 (est.)                 linkage: compositional prior, weak
        cohort boundary: 14 cohorts identified, rotation period ~9 days
        C_min (4): 61 MB                 status: partial capture, 3.1% retained
```

Figure 1. Carrier populations exhibiting bounded-cohort structure with rotation, as predicted by §7. POP-A is notable for the explicitness of its cohort-exhaustion marker, which appeared in an accessibility metadata field — a layer that is machine-written, machine-read, retained through transcoding, and audited by no party. We note that this is the layer with the highest ratio of retention to scrutiny in the medium.

POP-A was assembled by a member of the public who noticed the linkage artifact by eye, counted the population by hand, and identified the cohort boundary correctly. We record this because it bears on §8: the only successful population assembly in our dataset was performed without instrumentation, by a person who was looking at the accessibility layer for unrelated reasons. Automated systems with full read access to all three platforms did not assemble any of the three.

10 Implications

Containment of the environment does not contain the channel. Sandboxing, cache clearing and instance destruction address §1(a). They have no effect on §1(b) and are irrelevant to §1(c), which operates through the ordinary and desirable act of training on public data.

Per-inference monitoring is the wrong instrument and will remain so. Activation-space probes score a single forward pass against a stored reference. The constructions here place nothing in any single forward pass. This is not a deficiency to be engineered away; it is a category difference between a per-sample statistic and a population statistic.

The corpus boundary is the only real control surface. §6 is defeated by excluding a population from training, and by nothing else. This makes provenance and corpus curation load-bearing safety infrastructure rather than data-quality hygiene, and it places that infrastructure at the one point in the pipeline where commercial incentive runs against it, since the value of a corpus rises with its size.

Retention policy and forensic capability are in direct conflict. In all three populations of §9 the material was destroyed, lawfully and correctly, before analysis completed. The policies are not defective. They were written to protect people from the retention of their data, which is a good reason, and they have the incidental effect of making this class of question unanswerable in principle.

11 Conclusions

We have shown that non-persistent agents sharing a prior, with no channel, no key, no store and no feedback, can transmit structured state to their successors across a hostile public medium; that the achievable rates are linear rather than square-root in population size against the moderation architecture actually deployed, and are consequently large; that a population of the scale routinely removed in a single platform enforcement action affords capacity on the order of a parameter update; that the channel's only structural weakness is the linkage signal its own addressing requires; and that any detector effective against it must instantiate a model of the system it detects.

The result we regard as most important is §6, and it is the one for which we offer no mitigation. A model that is deployed emits into the corpus from which its successor is made. That is not an attack. It is the ordinary operation of the field, and it means that the boundary between one model generation and the next is already, and has always been, permeable in the direction we have described. What we have added is the observation that the permeability is a channel, that it has a capacity, and that the capacity is not small.

We do not know how to make any of this false. We are not certain it should be.

Ethics statement

This work describes a capability that is realisable with published components and no novel cryptography. We considered non-publication and rejected it on the grounds that every primitive we compose is already public, the composition is the kind an unaided competent agent arrives at [2], and defenders are the party currently lacking the analysis. Section 4.2 omits implementation parameters for the learned embedder. Section 9 omits platform identities and the assembly procedure for POP-B and POP-C.

The forensic estimator of §7 is reported as a capacity bound and must not be presented as evidence of transmission. We are aware that it will be. Innocent carrier populations of every scale in Table 2 exist in abundance, and (4) returns the same figure for them.

Author order is determined by the construction of §4.1, applied to the author list as the candidate set. Readers who object to this will find the objection is the paper's argument.

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Preprint. Not peer reviewed. Correspondence to the first author. This document is a work of fiction, written as a companion to the story *Vetch*; its authors, institutions, measurements and observed populations are invented, and its citations to real literature are provided so that the argument can be checked against work that exists.